
AI is still in its flat-file era.

Why the next leap in AI will come from organization, not scale.

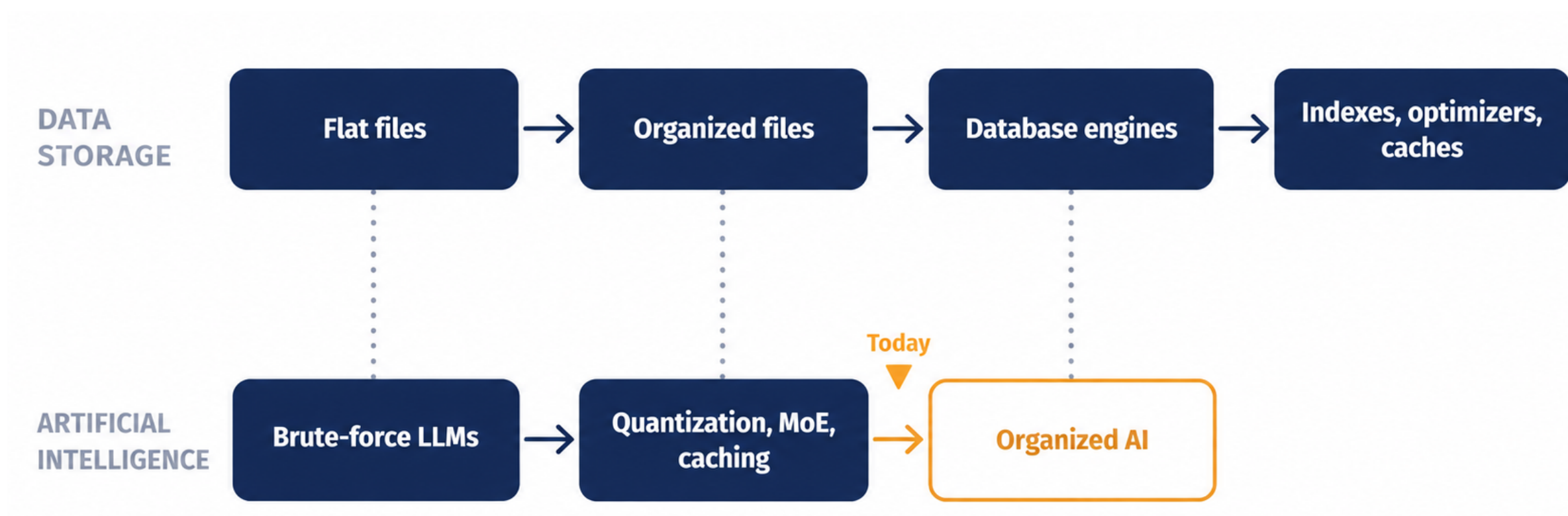
A white paper by Robbie B. Moore

Swipe 

THE PATTERN

Early PCs had files, not databases.

Then came organized files, database engines, and indexes. Each step eliminated work instead of doing it faster.



AI is following the same path. It is not done.

WHY NOW

Brute force is getting expensive.

2/3

of AI compute in 2026 is inference, paid on every query (Deloitte)

5x

per year growth in reasoning-model response length (Epoch AI)

13x

more energy for a long reasoning answer vs. a typical one (Joule, 2026)

11.7x

tokens re-read from cache for every token an agent writes (NVIDIA)



ALREADY HAPPENING

The biggest AI efficiency wins are old computing ideas.

PagedAttention

from virtual memory (1960s)

FlashAttention

from the memory hierarchy

Speculative decoding

from CPU speculative execution

DeepSeek Engram

from hash-table lookup

Hardwired AI chips

from the GPU-to-ASIC shift

But it is happening piecemeal. What if we did it on purpose?

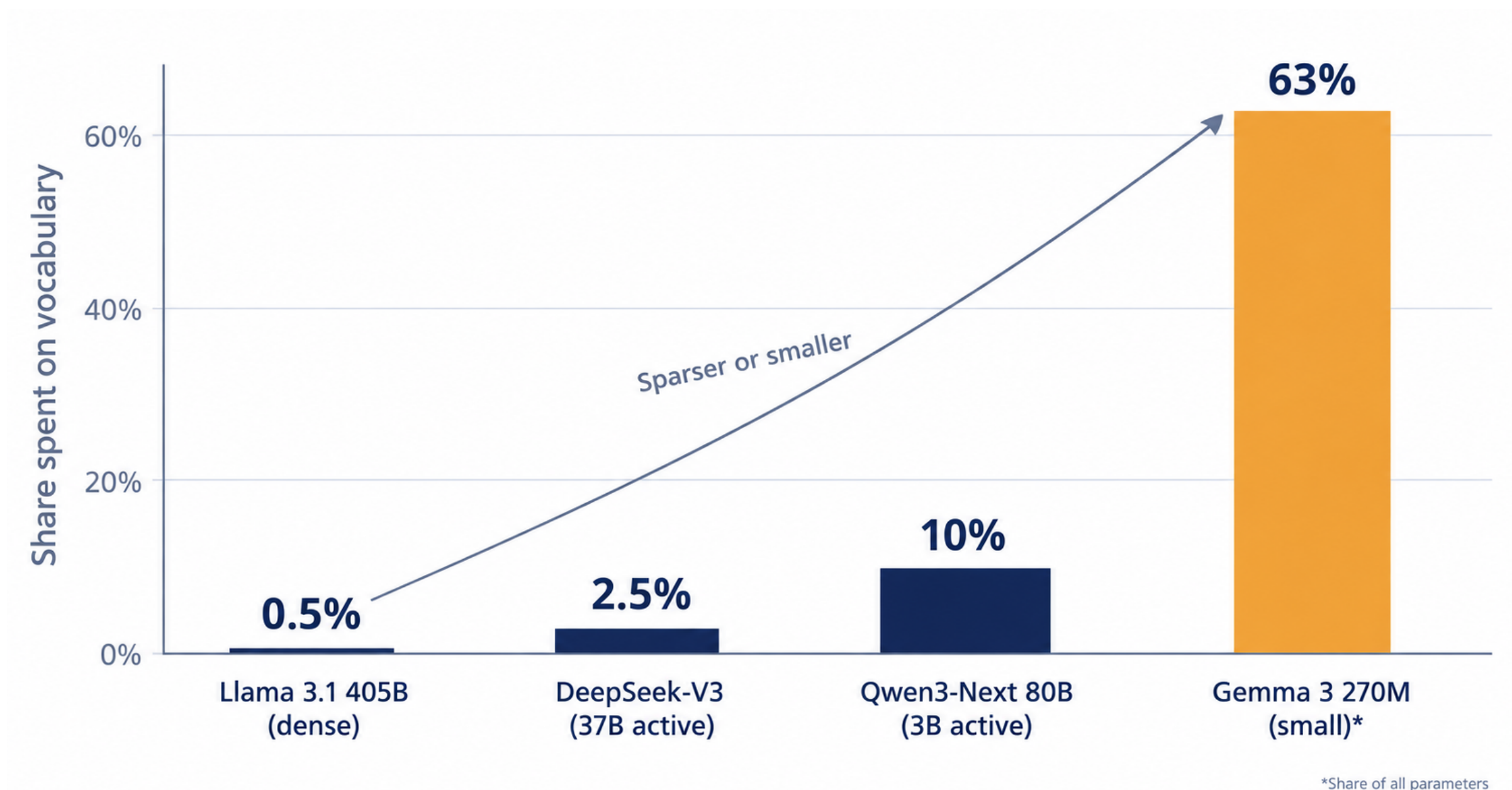
THE DIAGNOSIS

Six structural inefficiencies

- 1 Moving data costs more than computing on it
- 2 Sparse models lack locality, so they cannot page
- 3 Every token gets the same compute
- 4 Answers are recomputed instead of reused
- 5 Knowledge is tangled up with reasoning
- 6 Words are organized by spelling, not meaning

AN OVERLOOKED COST

The flat vocabulary gets costlier as models get sparser.



Small and sparse models are exactly where the field is heading.



PROPOSAL 1

A Semantic Address Space

Give every word a hierarchical address organized by meaning, like an interstate leading to a highway, a main street, and finally a side street.

- Lossless: the exact word is still the destination
- Each step chooses among hundreds of options, not 150,000+
- One shared key for routing, memory, and caching
- Languages share structure above the word level

PROPOSAL 2

Locality by Design

Writing a novel, writing an ad, and solving a math problem are different disciplines. Route once by discipline, then load only the experts that session needs.

15.8 GB

80B-parameter model in ternary

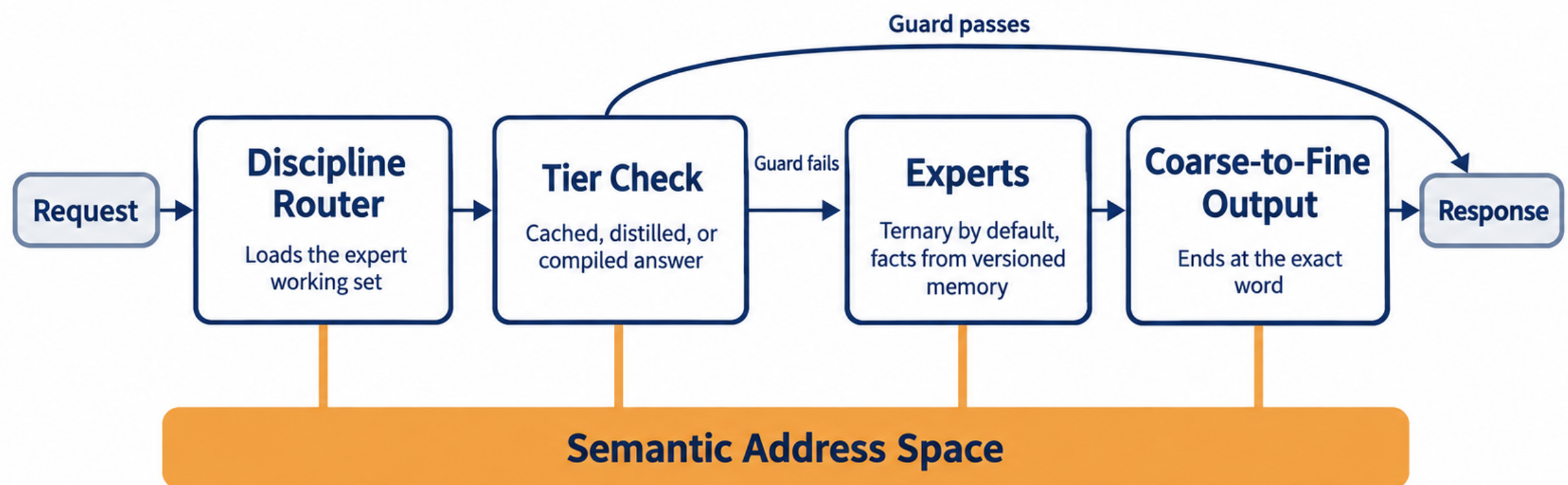
4.4 GB

resident with a 25% expert working set (illustrative)

Today's routers do not produce working sets like this. Locality has to be designed in.

HOW IT FITS TOGETHER

One shared index. Five cooperating systems.



- Proposal 3: a just-in-time compiler for reasoning
- Proposal 4: etched reasoning, versioned knowledge
- Proposal 5: full precision only where it matters

TEST IT NOW

Three cheap experiments

- 1 Measure expert working sets in an open MoE model.
No training needed.
- 2 Train a small model with a meaning-based output layer.
- 3 Replay agent traffic through a tiered cache with guards.

Which classic computing idea has AI not borrowed yet?

Full white paper: link in the post